To Work or to Shirk: The Shapiro-Stiglitz Theory

Shapiro, C.; Stiglitz, J.E. (1984). "Equilibrium Unemployment as a Worker Discipline Device". The American Economic Review. 74 (3): 433-444

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October 25, 2026

Motivation

- First point
- Second point

Worker's Choice

An individual worker's utility function is given by:

$$U(w, e) = w - e$$

where w is the wage and e is the effort level. Her lifetime utility is given by:

$$W = E \int_0^\infty u(w(t), e(t)) \exp(-rt)\, dt$$

where r is the discount rate.
She has the choice to either work or shirk. When working, she exerts effort e and receives wage w. When shirking, she exerts no effort and receives the same wage w. However, a worker who shirks faces a probability q of being caught and fired, in which case she receives unemployment benefits $\bar{w}$. Regardless of effort e, a worker also faces an exogenous probability b of being fired.

Asset Equations I

If she decides to work, her asset equation is given by:

$$rV_E^N = w - e + b(V_u - V_E^N)$$

- Probability of being fired: b (exogenous)
- Loss: $V_u - V_E^N$.

If she decides to shirk, her asset equation is given by:

$$rV_E^S = w + (b + q)(V_u - V_E^S)$$

- Probability of being fired: $b + q$ (exogenous + endogenous)
- Loss: $V_u - V_E^S$.

Asset Equations II

Finally, if she is unemployed, her asset equation is given by:

$$rV_u = \bar{w} + a(V_E - V_u)$$

- Unemployment benefits: $\bar{w}$
- Probability of being hired: a
- Gain: $V_E - V_u$.

No-Shirking Condition

In order for a worker to exert effort (not shirk), the following condition must hold:

$$V_E^N \geq V_E^S$$

Plugging in the asset equations, we get the **No-Shirking Condition (NSC)**:

$$w \geq rV_u + \frac{(r+b+q)e}{q} \equiv \hat{w}$$

whereby $\hat{w}$ is the *critical wage*.

Firm's Choice

- If $w < \hat{w}$, workers will shirk due to (1) the low wage w and (2) a high probability a of being rehired (high labor demand at low w). $\Rightarrow$ Firm should increase w to $\hat{w}$ to prevent shirking.
- If $w > \hat{w}$, workers will exert effort e due to (1) the high wage w and (2) a low probability a of being rehired (low labor demand at high w). $\Rightarrow$ Firm should decrease w to $\hat{w}$ to save costs.

Aggregate NSC

In steady state, the employment pool L stays constant, i.e. the flow into unemployment must equal the flow out of unemployment:

$$bL = a(N - L) \Leftrightarrow a = \frac{bL}{N - L}$$

where L is the number of employed workers and N is the total number of workers. Now plugging steady state a and V_u into the NSC, we get the **Aggregate NSC**:

$$w \geq e + \bar{w} + \frac{e}{q}\left(\frac{b}{u} + r\right) \equiv \hat{w} \quad \text{where } u = \frac{N - L}{N}$$

Pareto Efficiency

- First point
- Second point

Extensions

- First point
- Second point

Discussion

- First point
- Second point